J. Lesner

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Interests

Building machine-learning systems whose outputs are reliable enough to deploy in high-stakes, human-facing settings.

Education

University of California, Santa Barbara — **Santa Barbara, CA**
Ph.D. in Computer Science — *2026 – present*

- **Awarded:** Eugene Cota-Robles Fellowship

University of California, Santa Barbara — **Santa Barbara, CA**
M.S. in Computer Science, GPA 3.9/4.0 — *2024 – 2025*

- **Masters Project:** "Algorithmic Prompt Optimization for SQL Synthesis and Counterexample Discovery" **Advisor:** Dr. Xifeng Yan

University of California, Santa Cruz — **Santa Cruz, CA**
B.S. Computer Science with Highest Honors, GPA 3.9/4.0 — *2021 – 2024*

- Dean's Honors List and graduated with University Honor, Cum Laude

Publications

- **Lesner, J.**, & Beyeler, M. (2026). SymbolSight: Minimizing Inter-Symbol Interference for Reading with Prosthetic Vision. *IEEE EMBC 2026* (oral). [Video]
- **Lesner, J.**, Murayama, L., Guizar, T., Phunjamaneechot, P., & Shapiro, D. (2024). AI Personalized Interactive Fiction for Young Children. *PAIS-2024*, in *Frontiers in Artificial Intelligence and Applications: Vol. 392. ECAI 2024* (pp. 4756-4763). IOS Press.
- **Lesner, J.**, Murayama, L., Guizar, T., Phunjamaneechot, P., & Shapiro, D. (2024). A Demonstration of AI Personalized Interactive Fiction for Young Children. *ECAI-2024 demo track*, in *Frontiers in Artificial Intelligence and Applications: Vol. 392. ECAI 2024* (pp. 4487-4490). IOS Press. [Video]

Reports

- **Lesner, J.**, Zhao, F., & Yan, X. (2025). DBDoctor: LLM-Aided SMT Refutation of SQL Query Equivalence. *Technical Report*.
- **Lesner, J.** & Anand S. (2025). NutriGNN: Food Nutrient Prediction with an LLM Enriched Knowledge Graph. *Technical Report*.
- Dhaliwal, M. & **Lesner, J.**, (2025). MIRROR: Measuring, Improving, and Reproducing Ranking with Open Retrieval Models. *Technical Report*.
- Krachenfels, K. & **Lesner, J.** (2024). Understanding XAI Requirements: A Comparative Study of Repetitive and Unique Decision Contexts. *Technical Report*.
- **Lesner, J.**, & Elkaim, G. (2024). An Automated Tool for State Machine Diagram Generation in Mechatronics Education. *Technical Report*. [Video]

PROTOTYPES

SnipDue: AI Schedule Sync **SBHacks XI**
Hackathon project *2025*

- Built a schedule-synchronization app using Claude Sonnet and Cloudflare Workers to transform unstructured course information into calendar events (won "Best Use of Gen AI"). [Video]

GUARD: Guided Understanding & Agreement Rights Detector **UCSB**
Smartphone App User Interface in Figma *Fall 2024*

- Designed a prototype UI to detect and highlight troublesome or conflicting statements inside spoken and written communications.

AI Personalized Interactive and Teaching Fiction **UCSC**
Two generative AI powered Web Apps *2024*

- Built two web apps that use generative AI to entertain and teach young children using stories that have pictures, music, sounds, decisions and quizzes.

PRESENTATIONS

Oral talk: SymbolSight **Toronto, Canada**
IEEE EMBC 2026, session Sensory Neuroprostheses II *2026*

Retrospective on AIPIF / Project Team Mentoring **UCSC**
Annually invited speaker at CMPM146: Game AI Course *2024 – 2025*

SnipDue: AI-Powered Academic Deadline Management **UCSB**
AI CoP Spring Symposium 2025 *2025*

EXPERIENCE

CAHSI/NSF-Sponsored Research Assistant **UCSC**
Supervisor: Dr. Gabriel Elkaim *2022 – 2023*

- Engineered a static-analysis pipeline using XPATH/XSLT to automatically reconstruct state diagrams from diverse C-based embedded control code in mechatronics projects and authored a technical report detailing the static-analysis and visualization approach.

SERVICE

Women in Computer Science (WiCS), Graduate Officer **UCSB**
BUNS: Bunnies Urgently Needing Shelter, Volunteer **Santa Barbara, CA**
ASAP: Animal Shelter Assistance Program, Volunteer **Santa Barbara, CA**
Silicon Valley Bicycle Exchange, Volunteer bike mechanic **Palo Alto, CA**

SKILLS

AI/ML: DSPy, scikit-learn, PyTorch, TensorFlow, Jupyter notebooks
Languages: Python, Java, C, SQL, JavaScript/HTML, XSLT/XPath, LaTeX
Infrastructure: VSCode, Linux, Git, Docker, Databricks, Cloudflare